TCET/EXAM/FRM/IP-03/06 REVISION : B

IN-SEMESTER EXAMINATION-I

F.E./F.T. (Semester-I)

SUBJECT – Chemistry

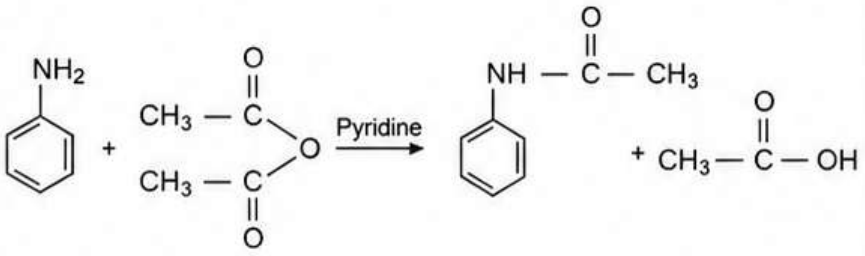
Branch: E&TC, E&CS, IT, MECH, M&ME, AI&ML
Div.: ALL
Duration: 60 Minutes

Date: 27/10/2025
Timing: 4:00 PM to 5:00 PM
Maximum Marks: 20

Instructions –

1. All questions are compulsory.
2. Assume suitable data wherever necessary and state the assumptions made.
3. Diagrams / sketches should be given wherever necessary.
4. Use of logarithmic table, drawing instruments and non-programmable calculators is permitted.
5. Figures to the right indicate full marks.

Q.1		Answer any 5 of the following questions	Marks	Course Outcomes	Learning Levels
	a.	Differentiate between temporary and permanent hardness.	2	CO1	U
	b.	Name any two common coagulants added to water with their chemical formulae.	2	CO1	A
	c.	Calculate total hardness of a water sample: $Ca(HCO_3)_2$ = 81 mg/L, $MgCl_2$ = 95 mg/L, $CaSO_4$ = 136 mg/L.	2	CO1	R
	d.	Mention any two advantages of composite materials compared to conventional materials.	2	CO2	U
	e.	What are nanomaterials. List out types of nanomaterials.	2	CO2	R
	f.	Discuss the green synthesis of disodium iminodiacetate.	2	CO3	A
	g.	Why should waste be prevented in chemical synthesis?	2	CO3	R
Q.2	a.	Calculate the quantity of pure lime (85% pure) and soda (95% pure) required for softening of 100,000 litres of water containing the following impurities in ppm. $Ca(HCO_3)_2$ = 81, $Mg(HCO_3)_2$ = 73, $CaCl_2$ = 22.2, $MgCl_2$ = 9.5, $MgSO_4$ = 6.7	5	CO1	A
		OR			
	b.	What is nanotube. Explain the precipitation method for synthesis of nanomaterials.	5	CO2	R
Q.3		Write the chemical formulae of zeolite, explain the principle, theory, reactions with suitable diagram for softening of hard water using zeolite method.	5	CO1	U
		OR			

	b.	Describe how renewable feedstock supports sustainable development. Calculate the % atom economy for the following reaction: (Atomic Weights: C = 12, N = 14, H = 1, O = 16)  	5	CO3	A